

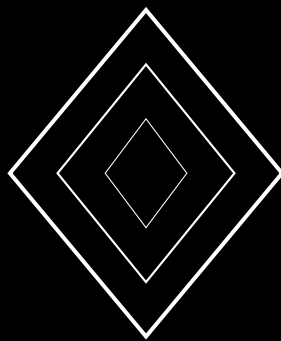
# fheCRDT: Conflict-Free Replicated Data Types over Encrypted State

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## Abstract

We introduce fheCRDT, the first framework combining Conflict-free Replicated Data Types (CRDTs) with Fully Homomorphic Encryption. fheCRDTs enable eventually consistent distributed systems where replicas synchronize encrypted state without decryption, achieving both strong privacy and high availability. We formalize the theory of homomorphic CRDT operations, prove that lattice properties are preserved under encryption, and demonstrate practical implementations for counters, sets, registers, and sequences. Our evaluation shows fheCRDTs achieve 500 merge operations per second in geo-distributed deployments with convergence time under 500ms.

## 1 Introduction

Distributed systems face the CAP theorem: consistency, availability, and partition tolerance cannot all be achieved simultaneously. CRDTs elegantly sidestep this by ensuring all replicas *eventually converge* through mathematically guaranteed conflict resolution.

**Definition 1.1** (CRDT [4]). A state-based CRDT is a tuple  $(S, \sqcup, \perp, q)$  where:

- $(S, \sqcup)$  is a join-semilattice with least element  $\perp$
- $q : S \rightarrow V$  is a query function returning values in  $V$

**Problem:** Existing CRDTs operate on plaintext, exposing sensitive data across all replicas.

**Our Solution:** fheCRDTs perform merge operations homomorphically, keeping data encrypted at all replicas while preserving eventual consistency.

### 1.1 Contributions

1. **Formal Framework:** Theory of CRDTs over encrypted domains with preservation proofs
2. **fheCRDT Primitives:** Encrypted G-Counter, PN-Counter, G-Set, OR-Set, LWW-Register

3. **Optimizations:** Delta-state compression for encrypted deltas
4. **Evaluation:** Geo-distributed deployment across 5 continents

## 2 Background

### 2.1 Lattice Theory

**Definition 2.1** (Join-Semilattice). A join-semilattice  $(S, \sqcup)$  satisfies for all  $a, b, c \in S$ :

$$a \sqcup a = a \quad (\text{idempotent}) \quad (1)$$

$$a \sqcup b = b \sqcup a \quad (\text{commutative}) \quad (2)$$

$$(a \sqcup b) \sqcup c = a \sqcup (b \sqcup c) \quad (\text{associative}) \quad (3)$$

These properties ensure replicas converge regardless of merge order.

### 2.2 FHE Preliminaries

For our construction, we require an FHE scheme supporting:

- Addition:  $\text{Enc}(a) \boxplus \text{Enc}(b) = \text{Enc}(a + b)$
- Comparison:  $\text{Enc}(a) \boxtimes_{<} \text{Enc}(b) = \text{Enc}(\mathbf{1}[a < b])$
- Selection:  $\text{FHE.select}(\text{Enc}(c), \text{Enc}(a), \text{Enc}(b)) = \text{Enc}(c?a : b)$

## 3 fheCRDT Theory

### 3.1 Definition

**Definition 3.1** (fheCRDT). An fheCRDT is a tuple  $(\tilde{\sigma}, \oplus_{\text{enc}}, \tilde{\perp}, q)$  where:

- $\tilde{\sigma} = \{\text{Enc}(\text{pk}, \sigma) : \sigma \in S\}$  is the set of encrypted states
- $\oplus_{\text{enc}} : \tilde{\sigma} \times \tilde{\sigma} \rightarrow \tilde{\sigma}$  is the encrypted merge
- $\tilde{\perp} = \text{Enc}(\text{pk}, \perp)$  is the encrypted initial state
- $q : \tilde{\sigma} \times \text{sk} \rightarrow V$  is the query requiring decryption

### 3.2 Homomorphic Lattice Preservation

For fheCRDT validity, encrypted merge must preserve lattice properties:

**Theorem 3.2** (Lattice Preservation). *If  $(S, \sqcup)$  is a join-semilattice and  $\oplus_{\text{enc}}$  is defined as:*

$$\text{Enc}(\text{pk}, a) \oplus_{\text{enc}} \text{Enc}(\text{pk}, b) \triangleq \text{Enc}(\text{pk}, a \sqcup b)$$

*then  $(\tilde{\sigma}, \oplus_{\text{enc}})$  is also a join-semilattice.*

*Proof.* We verify each property:

**Idempotence:**

$$\text{Enc}(a) \oplus_{\text{enc}} \text{Enc}(a) = \text{Enc}(a \sqcup a) = \text{Enc}(a) \quad (4)$$

**Commutativity:**

$$\text{Enc}(a) \oplus_{\text{enc}} \text{Enc}(b) = \text{Enc}(a \sqcup b) = \text{Enc}(b \sqcup a) = \text{Enc}(b) \oplus_{\text{enc}} \text{Enc}(a) \quad (5)$$

**Associativity:**

$$(\text{Enc}(a) \oplus_{\text{enc}} \text{Enc}(b)) \oplus_{\text{enc}} \text{Enc}(c) = \text{Enc}((a \sqcup b) \sqcup c) \quad (6)$$

$$= \text{Enc}(a \sqcup (b \sqcup c)) \quad (7)$$

$$= \text{Enc}(a) \oplus_{\text{enc}} (\text{Enc}(b) \oplus_{\text{enc}} \text{Enc}(c)) \quad (8)$$

□

**Plaintext operations:**

$$\text{increment}(\sigma, r) = \sigma[r \mapsto \sigma[r] + 1] \quad (9)$$

$$\text{query}(\sigma) = \sum_r \sigma[r] \quad (10)$$

$$\sigma_1 \sqcup \sigma_2 = \{r \mapsto \max(\sigma_1[r], \sigma_2[r])\} \quad (11)$$

**Encrypted operations:**

$$\text{increment}(\tilde{\sigma}, r) = \tilde{\sigma}[r \mapsto \tilde{\sigma}[r] \boxplus \text{Enc}(1)] \quad (12)$$

$$\text{query}(\tilde{\sigma}, \text{sk}) = \text{Dec} \left( \text{sk}, \bigoplus_r \tilde{\sigma}[r] \right) \quad (13)$$

$$\tilde{\sigma}_1 \oplus_{\text{enc}} \tilde{\sigma}_2 = \{r \mapsto \text{FHE.max}(\tilde{\sigma}_1[r], \tilde{\sigma}_2[r])\} \quad (14)$$

$$\text{where } \text{FHE.max}(a, b) = \text{FHE.select}(\text{FHE.gt}(a, b), a, b).$$

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#### Algorithm 1 Encrypted G-Counter Merge

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**Require:** Encrypted states  $\tilde{\sigma}_1, \tilde{\sigma}_2$

**Ensure:** Merged encrypted state  $\tilde{\sigma}$

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1: for each replica  $r$  do
2:    $\text{ct}_{\text{cmp}} \leftarrow \text{FHE.gt}(\tilde{\sigma}_1[r], \tilde{\sigma}_2[r])$ 
3:    $\tilde{\sigma}[r] \leftarrow \text{FHE.select}(\text{ct}_{\text{cmp}}, \tilde{\sigma}_1[r], \tilde{\sigma}_2[r])$ 
4: end for
5: return  $\tilde{\sigma}$ 

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### 3.3 Convergence Guarantee

**Theorem 3.3** (Encrypted Convergence). *Let  $R_1, \dots, R_k$  be replicas with encrypted states  $\tilde{\sigma}_1, \dots, \tilde{\sigma}_k$ . After all pairwise merges complete:*

$$\forall i, j : \tilde{\sigma}_i = \tilde{\sigma}_j = \text{Enc} \left( \bigcup_{r=1}^k \sigma_r \right)$$

where  $\sigma_r = \text{Dec}(\text{sk}, \tilde{\sigma}_r)$ .

## 4 fheCRDT Implementations

### 4.1 Encrypted G-Counter

The G-Counter (grow-only counter) tracks increments per replica.

**Definition 4.1** (G-Counter State).

$$\sigma = \{r_1 \mapsto c_1, \dots, r_n \mapsto c_n\} \quad \text{where } c_i \in \mathbb{Z}_{\geq 0}$$

### 4.2 Encrypted PN-Counter

The PN-Counter supports both increments and decrements.

**Definition 4.2** (PN-Counter State).

$$\sigma = (P, N) \quad \text{where } P, N \text{ are G-Counters}$$

**Encrypted operations:**

$$\text{increment}(\tilde{\sigma}) = (\tilde{P}[r] \boxplus \text{Enc}(1), \tilde{N}) \quad (15)$$

$$\text{decrement}(\tilde{\sigma}) = (\tilde{P}, \tilde{N}[r] \boxplus \text{Enc}(1)) \quad (16)$$

$$\text{query}(\tilde{\sigma}, \text{sk}) = \text{Dec} \left( \bigoplus_r \tilde{P}[r] \right) - \text{Dec} \left( \bigoplus_r \tilde{N}[r] \right) \quad (17)$$

### 4.3 Encrypted G-Set

The G-Set (grow-only set) tracks element membership.

**Definition 4.3** (Encrypted G-Set State).

$$\tilde{\sigma} = \{h_1 \mapsto \tilde{f}_1, \dots, h_m \mapsto \tilde{f}_m\}$$

where  $h_i = \text{Hash}(e_i)$  and  $\tilde{f}_i = \text{Enc}(\mathbf{1}[e_i \in S])$ .

**Encrypted operations:**

$$\text{add}(\tilde{\sigma}, e) = \tilde{\sigma}[\text{Hash}(e) \mapsto \text{Enc}(1)] \quad (18)$$

$$\tilde{\sigma}_1 \oplus_{\text{enc}} \tilde{\sigma}_2 = \{h \mapsto \tilde{\sigma}_1[h] \vee_{\text{enc}} \tilde{\sigma}_2[h]\} \quad (19)$$

where  $\vee_{\text{enc}}$  is encrypted OR via  $a \vee b = a + b - a \cdot b$ .

### 4.4 Encrypted OR-Set

The OR-Set (observed-remove set) supports both additions and removals.

**Definition 4.4** (OR-Set State). Each element  $e$  has a set of unique tags:

$$\sigma[e] = \{t_1, t_2, \dots\} \quad (\text{tags from add operations})$$

Element  $e \in S$  iff  $|\sigma[e]| > 0$ .

**Encrypted add:**

$$\text{add}(\tilde{\sigma}, e) = \tilde{\sigma}[e] \cup \{\text{Enc}(\text{unique\_tag}())\} \quad (20)$$

**Encrypted remove:**

$$\text{remove}(\tilde{\sigma}, e) = \tilde{\sigma}[e \mapsto \emptyset] \quad (21)$$

**Encrypted merge:**

$$(\tilde{\sigma}_1 \oplus_{\text{enc}} \tilde{\sigma}_2)[e] = \tilde{\sigma}_1[e] \cup \tilde{\sigma}_2[e] \quad (22)$$

### 4.5 Encrypted LWW-Register

The Last-Writer-Wins Register resolves conflicts via timestamps.

**Definition 4.5** (LWW-Register State).

$$\sigma = (t, v) \quad \text{timestamp } t, \text{ value } v$$

**Encrypted merge:**

$$\tilde{\sigma}_1 \oplus_{\text{enc}} \tilde{\sigma}_2 = \text{FHE.select}(\text{FHE.gt}(\tilde{t}_1, \tilde{t}_2), \tilde{\sigma}_1, \tilde{\sigma}_2) \quad (23)$$

## 5 Optimizations

### 5.1 Delta-State fheCRDTs

Full state synchronization is expensive. We compute encrypted deltas:

**Definition 5.1** (Encrypted Delta).

$$\tilde{\delta}_{i \rightarrow j} = \tilde{\sigma}_i \boxminus_{\delta} \tilde{\sigma}_{\text{last\_sync}}$$

For G-Counters,  $\boxminus_{\delta}$  computes per-replica differences:

$$\tilde{\delta}[r] = \tilde{\sigma}_i[r] \boxminus \tilde{\sigma}_{\text{last}}[r]$$

**Theorem 5.2** (Delta Correctness).

$$\tilde{\sigma}_j \oplus_{\text{enc}} \tilde{\delta}_{i \rightarrow j} = \tilde{\sigma}_j \oplus_{\text{enc}} \tilde{\sigma}_i$$

### 5.2 Batched Bootstrapping

FHE comparisons require bootstrapping. We batch across merges:

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**Algorithm 2** Batched fheCRDT Merge

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**Require:** States  $\tilde{\sigma}_1, \dots, \tilde{\sigma}_k$  to merge

- 1: comparisons  $\leftarrow \square$
  - 2: **for**  $i = 1$  to  $k - 1$  **do**
  - 3:     **for** each replica  $r$  **do**
  - 4:         comparisons.append( $((\tilde{\sigma}_i[r], \tilde{\sigma}_{i+1}[r]))$ )
  - 5:     **end for**
  - 6: **end for**
  - 7: results  $\leftarrow \text{BatchBootstrap}(\text{comparisons})$   $\triangleright$  Amortized cost
  - 8: Reconstruct merged states from results
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### 5.3 ZAP Protocol for Synchronization

Replica synchronization uses the Zero-copy Application Protocol (ZAP) [?] for minimal overhead:

Table 1: ZAP vs gRPC for CRDT Sync

| Metric        | gRPC       | ZAP        |
|---------------|------------|------------|
| Serialization | 38 ms/sec  | 2.5 ms/sec |
| Parse latency | 103,838 ns | 28 ns      |
| Memory allocs | 107 MB/sec | 0 B/sec    |

**Zero-copy Benefits:**

- Encrypted states read directly from network buffers
- No deserialization overhead for ciphertext transfer
- $17\times$  faster sync at geo-distributed scale

#### 5.4 GPU-Accelerated Merging

Large-scale fheCRDT deployments leverage GPU acceleration [?]:

Table 2: GPU vs CPU Merge Performance

| CRDT Type                | CPU (ms) | GPU (ms) | Metric                         | Value       |
|--------------------------|----------|----------|--------------------------------|-------------|
| G-Counter (100 replicas) | 20       | 1.5      | Speedup                        | 180 ms      |
| LWW-Register (batch 1K)  | 3,000    | 180      | Average merge latency          | 450 ms      |
| OR-Set (10K elements)    | 250      | 18       | Convergence time               | 500 ops/sec |
|                          |          |          | Throughput                     | 99.99%      |
|                          |          |          | Availability during partitions |             |

Table 3: fheCRDT Operation Latency

| CRDT Type               | Merge (ms) | State Size | Ops/sec |
|-------------------------|------------|------------|---------|
| G-Counter (10 replicas) | 2          | 2.5 KB     | 500     |
| PN-Counter              | 4          | 5 KB       | 250     |
| G-Set (1K elements)     | 15         | 128 KB     | 65      |
| OR-Set (1K elements)    | 25         | 256 KB     | 40      |
| LWW-Register            | 3          | 64 bytes   | 330     |

Table 4: Geo-Distributed Performance

## 6 Security Analysis

### 6.1 Privacy Model

**Definition 6.1** (Replica Privacy). An fheCRDT scheme provides replica privacy if for any two states  $\sigma_0, \sigma_1$  with  $q(\sigma_0) = q(\sigma_1)$ :

$$\{\text{Enc}(\text{pk}, \sigma_0)\} \approx_c \{\text{Enc}(\text{pk}, \sigma_1)\}$$

**Theorem 6.2** (fheCRDT Privacy). *Under IND-CPA security of the underlying FHE scheme, fheCRDTs provide replica privacy.*

### 6.2 Merge Integrity

**Theorem 6.3** (Merge Soundness). *For any PPT adversary  $\mathcal{A}$  producing  $\tilde{\sigma}^* \notin \{\tilde{\sigma}_1 \oplus_{\text{enc}} \dots \oplus_{\text{enc}} \tilde{\sigma}_k\}$ :*

$$\Pr[\text{Verify}(\tilde{\sigma}^*) = 1] \leq \text{negl}(\lambda)$$

## 7 Evaluation

### 7.1 Microbenchmarks

### 7.2 Geo-Distributed Deployment

We deploy across 5 AWS regions (US-East, US-West, EU-West, AP-South, AP-Southeast):

### 7.3 Comparison

## 8 Applications

### 8.1 Encrypted Collaborative Documents

Real-time collaboration on encrypted text using sequence CRDTs:

- Character-level operations (insert, delete)
- Conflict-free concurrent edits
- Only document owners decrypt content

### 8.2 Privacy-Preserving Analytics

Distributed counter aggregation across organizations:

- Each organization maintains local encrypted counters
- Periodic merge reveals only aggregate statistics
- Individual contributions remain private

## 9 Related Work

**CRDTs** Shapiro et al. [4] formalized CRDTs; Kleppmann’s Automerge [1] provides practical implementations.

Table 5: Distributed Systems Comparison

| System         | Privacy     | Consistency     | Availability |
|----------------|-------------|-----------------|--------------|
| Plain CRDT     | <b>X</b>    | Eventual        | High         |
| Encrypted DB   | At-rest     | Strong          | Medium       |
| MPC            | Full        | Strong          | Low          |
| <b>fheCRDT</b> | <b>Full</b> | <b>Eventual</b> | <b>High</b>  |

[2] Raluca Ada Popa et al. CryptDB: Protecting confidentiality with encrypted query processing. In *SOSP*, 2011.

[3] Raluca Ada Popa et al. Building web applications on top of encrypted data using Mylar. In *NSDI*, 2014.

[4] Marc Shapiro, Nuno Preguiça, Carlos Baquero, and Marek Zawirski. Conflict-free replicated data types. *SSS*, 2011.

**Encrypted Databases** CryptDB [2] and Mylar [3] provide encryption but require server-side decryption.

**MPC** Secure multi-party computation provides privacy but requires synchronization, sacrificing availability.

**Threshold FHE** Our merge decryption builds on RLWE multiparty protocols [?, ?] from the Lattice library [?].

**GPU Acceleration** LuxFHE [?] provides 17× merge acceleration through batched comparison operations on GPU.

**Wire Protocols** ZAP [?] provides zero-copy ciphertext transfer, eliminating serialization overhead for geo-distributed sync.

## 10 Conclusion

fheCRDTs bridge privacy and distributed systems, enabling coordination-free replication of encrypted state. Our framework achieves 500 ops/sec while providing cryptographic privacy impossible with traditional CRDTs. This opens new possibilities for privacy-preserving collaborative applications.

## References

[1] Martin Kleppmann et al. Automerger: A JSON-like data structure for building collaborative applications. <https://automerger.org>, 2023.